SafeDrive Insurance Ltd.

Predicting Policyholder Claims: Business Report

Audience: Underwriting, Pricing and Risk Management leadership | Companion file: dashboard_report.pdf

1. Executive Summary

SafeDrive filed **3,748 claims on 58,592 policies (6.40%)**. We built and compared two models to flag policyholders likely to claim. The selected model is a **tuned Decision Tree (test ROC-AUC 0.646, recall 65.3%, precision 8.9%)**, which beat the Logistic Regression baseline (ROC-AUC 0.585). The 761 policies in its top risk band claimed at **14.98%**, about 2.3 times the portfolio rate and 3.7 times the Low band (4.09%).

The signal is real but modest. The model is suitable as a screening aid that prioritises policies for review, not as an automatic accept/decline rule. It is not approved for production use in its current form.

2. Problem Statement, Context and Success Metrics

SafeDrive has seen rising claim payouts over the last two years. Manual underwriting is inconsistent, so management wants to identify policyholders likely to file a claim during the policy period, and to understand in plain language what drives that risk, so that underwriting and risk teams can act early.

Business questions

- Which policyholder, vehicle and policy characteristics are associated with a higher chance of claiming?
- Do vehicle safety attributes (NCAP rating, speed alert) and the engineered high-risk driver profile separate claimers from non-claimers?
- Can a model rank policies into risk bands that management can act on, and how reliable are those bands?

Success metrics

Type	Metric	Why it matters here
Business	Claim rate in the top risk band vs portfolio	Shows whether flagged policies truly claim more
Business	Share of actual claims captured (recall)	Missed claimers are the costly error for an insurer
Business	Review workload (policies flagged)	Keeps underwriting effort affordable
Model	ROC-AUC	Threshold-free ranking quality on a 6.4% claim base
Model	Recall and Precision / F1	Accuracy is misleading when 93.6% of policies do not claim

3. Data Summary and Preprocessing

The dataset has 58,592 policies and 44 columns (policy, policyholder, vehicle and safety attributes plus the target is_claim). There were no missing values, no duplicate rows and no duplicate policy_id values, so each row is an independent policy. The target is imbalanced: 3,748 claims against 54,844 non-claims (about 1 : 14.6).

Numerical features (selected)

Feature	Mean	Std	Min	Median	Max
policy_tenure (years)	0.61	0.41	0.00	0.57	1.40
age_of_car (normalised)	0.07	0.06	0.00	0.06	1.00
age_of_policyholder (normalised)	0.47	0.12	0.29	0.45	1.00
population_density (per sq.km)	18,827	17,660	290	8,794	73,430
displacement (cc)	1,162	266	796	1,197	1,498
gross_weight (kg)	1,385	212	1,051	1,335	1,720
ncap_rating (0-5)	1.76	1.39	0	2	5

Categorical features (selected, share of policies)

Feature	Distribution
segment	A 29.6%, B2 31.3%, C2 23.9%, B1 7.1%, C1 6.1%, Utility 2.1%
fuel_type	Petrol 35.0%, CNG 34.7%, Diesel 30.3%
transmission_type	Manual 65.2%, Automatic 34.8%
is_speed_alert	Yes 99.4% (58,229), No 0.6% (363)
area_cluster	22 clusters; C8 is largest (13,654 policies, 23.3%)

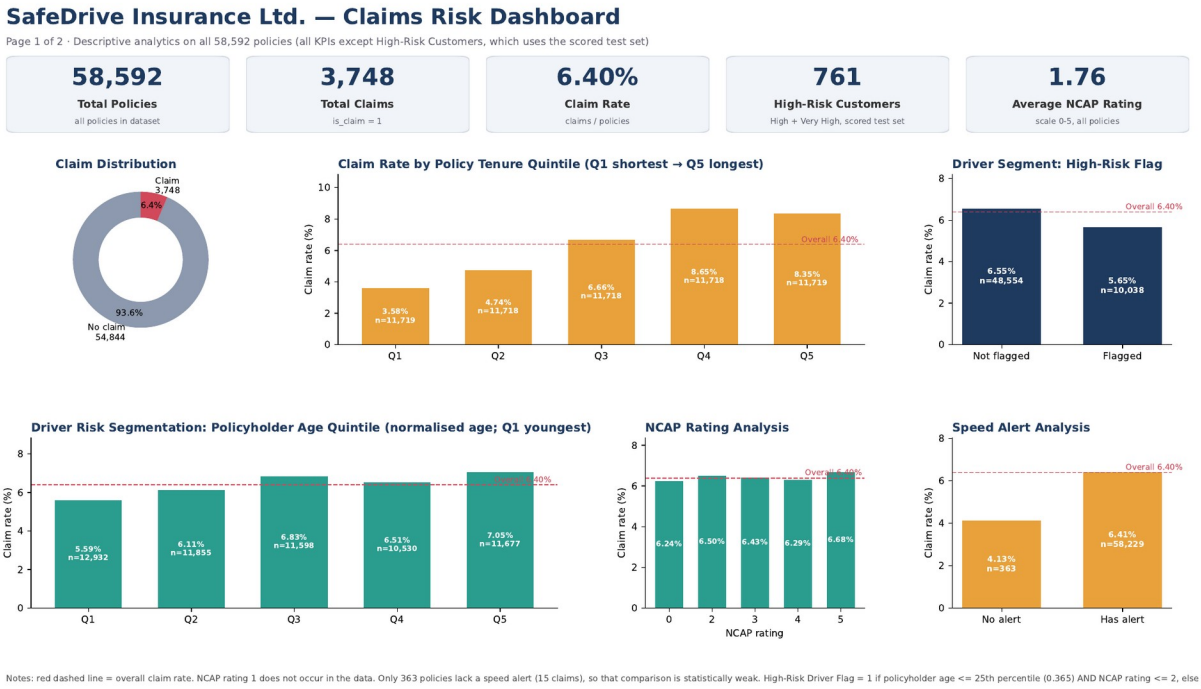
Preprocessing overview

- Yes/No flags were standardised, then encoded to 1/0; max_power and max_torque text (for example 88.50bhp@6000rpm) were parsed into numeric columns.
- Numeric outliers were capped with the IQR rule (Q1 - 1.5 IQR to Q3 + 1.5 IQR); nominal variables were one-hot encoded.
- An 80/20 stratified split (46,873 train, 11,719 test, random_state 42) kept the claim rate the same in both parts. Scalers and encoders were fitted on training data only to prevent leakage.

4. Key EDA Findings

- **Policy tenure is the clearest pattern.** Claim rate climbs from 3.58% in the shortest-tenure quintile to 8.65% in the fourth and 8.35% in the longest.
- **Vehicle age:** the oldest quintile of cars claims least (5.04%) versus 6.94% for the newest. Policyholder age shows a mild rise from 5.59% (lowest quintile) to 7.05% (highest) on the normalised age scale.
- **Geography matters.** Area-cluster claim rates range from 3.86% (C17) to 10.74% (C18, only 242 policies), so small clusters need cautious reading.
- **NCAP rating shows almost no separation:** rates sit between 6.24% and 6.68% across ratings, so safety stars alone do not explain claims here. No policy has a rating of 1.
- **Speed alert:** policies without one claim less (4.13%) than those with one (6.41%), but only 363 policies (15 claims) lack it, so this is not reliable evidence.
- **Segment, fuel and transmission** are close to the overall rate (for example segment B2 6.86% vs C1 6.41%; Petrol 6.64% vs CNG 6.07%).

- **Correlations with the claim flag are weak:** the largest absolute value is policy_tenure at 0.079, followed by age_of_car at 0.028. This foreshadows the model ceiling.



Notes: red dashed line = overall claim rate. NCAP rating 1 does not occur in the data. Only 363 policies lack a speed alert (15 claims), so that comparison is statistically weak. High-Risk Driver Flag = 1 if policyholder age <= 25th percentile (0.365) AND NCAP rating <= 2, else 0.

5. Feature Engineering and Feature Selection

Engineered features and business relevance

Feature	Definition	Business relevance
High_Risk_Driver_Flag	1 if policyholder age <= 25th percentile (0.365) AND NCAP rating <= 2, else 0	Encodes a classic underwriting profile (younger driver in a less safe car) in one rule
safety_equipment_count	Sum of binary safety-feature flags	Summarises how well-equipped the vehicle is; later removed as redundant
power_to_weight_ratio	max_power_bhp / gross_weight	Proxy for vehicle performance, which can relate to driving behaviour

Honest finding: the flagged group (10,038 policies) claimed at **5.65%**, slightly below the unflagged 6.55%. The flag as defined does not isolate riskier drivers in this data and should be redefined before it is used in pricing.

Why features were dropped

- **policy_id:** an identifier with no predictive meaning.
- **Raw max_power / max_torque text:** replaced by the parsed numeric columns.
- **17 near-duplicate columns (|correlation| > 0.9 with a retained column):** cylinder, gear_box, length, width, max_power_bhp, max_torque_nm, power_rpm, safety_equipment_count, and nine equipment flags (is_esc, is_brake_assist, is_central_locking, is_driver_seat_height_adjustable, is_ecw, is_front_fog_lights, is_rear_window_defogger, is_rear_window_washer, is_rear_window_wiper). Keeping them would add instability without new information.

- The remaining numeric features were checked with Variance Inflation Factor (VIF > 10 flagged). Many vehicle-specification variables still showed high VIF after pruning and were kept for model performance; this is noted as a limitation because it makes individual logistic coefficients less stable.

The final model uses 30 features: 22 numeric/binary and 8 categorical.

6. Model Results

Logistic Regression assumptions were checked. Independence holds (unique policy IDs, no duplicates). The Box-Tidwell test flagged non-linear log-odds for policy_tenure and population_density ($p < 0.05$), but not for age_of_car, height or age_of_policyholder. This supports trying a tree model, which does not need linearity.

Metric (test set)	Logistic Regression	Tuned Decision Tree
Accuracy	0.569	0.551
Precision	0.080	0.089
Recall	0.548	0.653
F1	0.140	0.157
ROC-AUC	0.585	0.646
5-fold CV ROC-AUC	0.610 ± 0.006	0.638 ± 0.013
5-fold CV F1	0.150 ± 0.003	0.158 ± 0.004

Selected model: Decision Tree (class_weight balanced; Optuna, 25 trials, TPE seed 42, 3-fold CV ROC-AUC). Best parameters: max_depth 4, min_samples_split 4, min_samples_leaf 3. It wins on test ROC-AUC (0.646 vs 0.585) and on cross-validation, and it catches more claimers (recall 65.3%). Accuracy is lower because the model is deliberately tuned to find claimers rather than to be right on the non-claimers.

The AutoML pass with Featuretools (multiply and add features) found no generated feature whose correlation with claims beat the best base feature, so zero were retained. The AutoML model is therefore the same as the Stage 2 model, with identical metrics and identical risk bands.

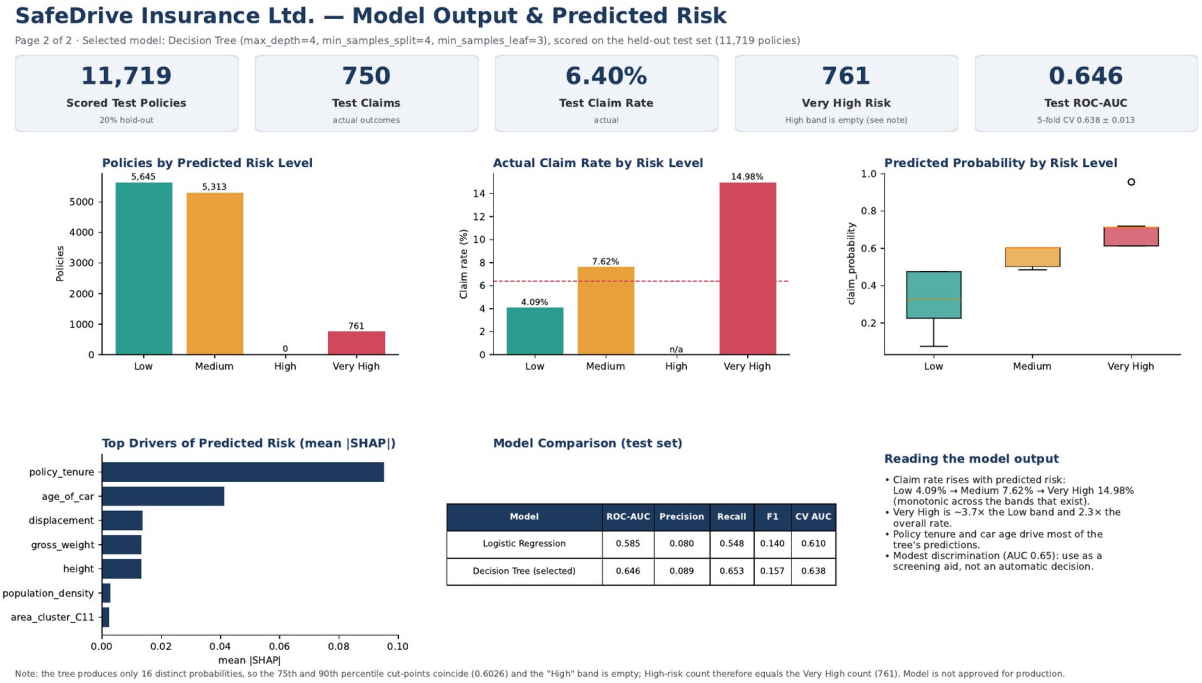
What drives predictions

- **policy_tenure** is the dominant driver (mean |SHAP| 0.095), followed by **age_of_car** (0.041). Displacement, gross weight and height contribute far less (about 0.013 each).
- In the baseline logistic model, longer tenure raises claim odds (odds ratio 1.47 per standard deviation), and area cluster C18 has the highest odds (2.30) while C17 is lowest (0.63).

Predicted risk bands (held-out test set, 11,719 policies)

Risk level	Policies	Actual claim rate	Mean predicted probability
Low	5,645	4.09%	0.339
Medium	5,313	7.62%	0.569
High	0	n/a	n/a
Very High	761	14.98%	0.687

Claim rates rise from Low to Medium to Very High, so the ordering is monotonic across the bands that exist. The tree outputs only 16 distinct probability values, so the 75th and 90th percentile cut-points are identical (0.6026) and the High band is empty. As a result, the high-risk customer count (High + Very High) is 761, all of them Very High.



7. Business Implications and Recommendations

- **Use the model to prioritise review, not to decide.** The 761 Very High policies claim at 14.98%; refer them for manual underwriting checks. Because precision is only 8.9%, roughly 9 in 10 flagged policies at the model's default threshold will not claim, so do not decline on the score alone.
- **Review pricing for longer-tenure policies.** Claim rates in the two longest-tenure quintiles (8.65% and 8.35%) are more than double the shortest (3.58%). Test renewal-time risk checks or a tenure-linked loading, after confirming what tenure represents in the business.
- **Monitor high-claim geographies.** Clusters such as C18 (10.74%), C22 (8.21%) and C14 (7.68%) merit monitoring and loss-ratio review; treat the smallest clusters with caution due to low volume.
- **Do not rely on NCAP or speed-alert status as pricing levers yet.** The data shows no meaningful difference by NCAP rating, and the speed-alert group without the feature is too small to judge.
- **Redesign the High-Risk Driver Flag** before using it operationally, since it currently marks a group that claims slightly less than average.
- **Monitor the model.** Track claim rate by band each quarter and re-train when the Very High band no longer outperforms the Low band.

8. Limitations and Next Steps

- Modest predictive power: ROC-AUC 0.646 and precision 8.9% mean many false alarms. The model is registered as not approved for production.
- Risk banding is coarse: the shallow tree gives only 16 distinct scores and no High band. A smoother model (gradient boosting, calibrated probabilities) would allow four useful bands.

- Policyholder and car ages are normalised, and the dataset lacks driver-behaviour fields (driving experience, past accidents, mileage, credit score) that typically carry stronger signal.
- Residual multicollinearity among vehicle-specification features; logistic coefficients should be read with caution. Small clusters and the 363 no-speed-alert policies give unstable rates.
- Associations are not causal, and results come from a single train/test split plus cross-validation.

Next steps: add behavioural data, try boosted trees with probability calibration, set the decision threshold using claim cost vs review cost, and run a controlled pilot of the referral process.